CURRICULUM VITAE

DR. SUNDARAMURTHY PANDURANGAN, PH.D.

https://www.linkedin.com/in/dr-sundaramurthy-pandurangan-30015827/

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Address for Communication
No. 5/132, Middle Street,
Govindapuram,
Venkatasamuthram,Ambur,
Vellore District**,** Tamil Nadu
PIN-635811, India

E - Mail
pksmurthy2020@gmail.com
Contact Number
091 - 9600404797
Citizenship
Indian

ACADEMIC RECORDS

Jan 2004 – July 2009 **Ph.D., Mathematics,**
Indian Institute of Technology Roorkee.

July 1993 – June 1995 **M.Sc., Mathematics,**
Sacred Heart College, University of Madras.

July1990 – June1993 **B.Sc., Mathematics,**
MazharulUloom College, University of Madras.

ACADEMIC AWARDS

March 2003 **Graduate Aptitude Test in Engineering (GATE) 2003,**
Score:*96.13Percentile,*All India Rank: *50*,
Indian Institute of Technology Madras.

1994 – 1995 **First Prize*for "Proficiency in Studies"*,**
Sacred Heart College, Tirupattur.

1992 – 1993 **First Prize*for"Proficiency in Mathematics"*,**
MazharulUloom College, Ambur.

PROFESSIONAL POSITIONS

June2024 –Oct2024 **Chief AI Scientist,**
WOU AI Research Center, Hyderabad.

Dec2021 – March2024 **Deep Learning Specialist,**
Standard Chartered Bank, Chennai.

Dec2020 – March2021 **Head of Data Science and AI,**
Pink Tech Design, Delhi.

June2018–Nov2020 **Professor of Mathematics / AI,**
Alliance University, Bangalore.

Feb2015 – Feb2018 **Subject Matter Expert in Molecular Biomarkers,**
PointCross Life Sciences, Bangalore.

Mar2012–Feb2014 **Professor of Mathematics / Network Biology,**
Vellore Institute of Technology, Vellore.

Aug2010–April 2011 **Post Doctoral Fellow in Computational Genomics,**
Cold Spring Harbor Laboratory, New York.

Aug2009–Nov2009 **Assistant Professor of Bioinformatics,**
Karunya Institute of Technology and Sciences, Coimbatore.

PROFESSIONAL SERVICES

22nd Aug 2024 **Session Chair,**
Artificial Intelligence and Knowledge Processing (AIKP'2024) Conference

11thOct-10th Nov 2021 **Data Science and AI Consultant,**
Revenue Box Technologies, Bangalore.

03rd-21st Aug 2020 **Resource Person (All Sessions),**
Workshop on Probability and Statistics with R,
Alliance University, Bangalore.

12th-28th June 2019 **Resource Person (All Sessions),**
Workshop on Engineering Mathematics with MATLAB,
Alliance University, Bangalore.

26th April 2019 **Session Chair,**
Alliance International Conference on Artificial Intelligence and Machine Learning (AICAAM 2019**),** Bangalore.

16th Dec 2017 **Member, IAAB (Industry/Alumni Advisory Board) ,**
Department of Computer Applications,
Madanapalle Institute of Technology and Science.

28th -29th Jan 2010 **Organizing Secretary,**
National Conference on Proteomic Applications: Perspectives in Healthcare (NCPPH), Karunya Institute of Technology and Sciences, Coimbatore.

2006 - 2007 **Member, Expert Committee to Redesign Curriculum,**
Department of Mathematics, Sacred Heart College, Tirupattur.

DATA SCIENCE/AI SKILLS

Agentic AI
Generative AI
Conversational AI
Document AI
Large Language Models
Multimodal Models
Forecasting Models
Deep Learning
Reinforcement Learning
Machine Learning
Probability and Statistics
Linear Algebra
Multivariate Calculus
Graph Theory

COMPUTING SKILLS

Python, R, MATLAB

SOFTWARE / AI PROJECTS

1. Salman et al**., Unified AI system for imagery-based disease diagnosis and prognosis,** WOU AI Research Center, 2024.

2. Pramodh et al., **Unified AI system for speech technology**, WOU AI Research Center, 2024.

3. Sundaramurthy P et al., **Integrated platform for psychometry**, WOU AI Research Center, 2024.

4. Vineeth K et al., **Unified AI system for time series forecasting**, WOU AI Research Center, 2024.

5. Sundaramurthy P et al., I**ntelligent document processing system,** Standard Chartered Bank, 2023.

6. Sundaramurthy P et al., **LLM-powered QA chatbot,** Standard Chartered Bank, 2022.

7. Sundaramurthy P, **Prototypes for deep learning model-based text classification, summarization and generation.** Standard Chartered Bank, 2022.

8. Sundaramurthy P, **Prototypes for recommender system**, RevenueBox Technologies, 2021.

9. Sundaramurthy P et al., **Computer vision based AI functionalities for national security**, Pink Tech Design, 2021.

10. Sundaramurthy P et al., **Xbiom Insights-Software product to integrate, search and analyze pharma R & D data** ,PointCross Life Sciences, 2017.
http://www.pointcrosslifesciences.com/index.php/clinical-research-development/

11. Sundaramurthy P et al., **HORIBALFRE-Higher Order Residue Interactions Based ALgorithm for Fold Recognition,** Presented at ICCABS 2011 and published in Bioinformation, 2011

12. Sundaramurthy P et al., **HORI - Web server to compute Higher Order Residue Interactions in proteins,** Presented at APBC 2010 and published in BMC Bioinformatics, 2010.

ACADEMIC PROJECTS

1. Mathew C et al., **Insights into housing sector using regression models**, Technical report submitted to Alliance University, 2020.

2. Bhoomkia A P et al., **Implementation of classification algorithms and their comparison for Irish dataset,**Technical report submitted to Alliance University, 2020.

3. Vijayalakshmi R et al., **Implementation of decision tree based predictive models,** Technical report submitted to Alliance University, 2020

4. Radha R et al., **Tree based machine learning models using python,** Technical report submitted to Alliance University, 2020

5. Bhandare T V et al., **Classifications using MLP, CNN and GAN,** Technical report submitted to Alliance University, 2020.

6. Disha S W et al., **Implementation of MLP, RNN and GAN algorithms,**Technical report submitted to Alliance University, 2020.

7. Pavithra M et al., **Exploratory and predictive analysis of the performance of posts posted in a product company's Facebook page**, Project report submitted to Analytics Training Institute, 2018.

8. MD Aksam VK, **Computational identification and analysis of cancer drug targets in the network of MAPK pathways** , Ph.D. thesis submitted to Vellore Institute of Technology, 2018.

9. MD Aksam VK et al., **Neural network based prediction of less side effects causing cancer drug targets in the network of MAPK pathways** ,Presented at Inbix'17 and published in IJBRA, 2017.

10. Sundaramurthy P, **Prediction and analysis of the dynamical behaviors of ERK5, JNK, and P38 kinase cascades**, Ph.D. thesis submitted to Indian Institute of Technology Roorkee, 2009.

RESEARCH PUBLICATIONS

1.MD Aksam V.K, Chandrasekaran V. M., and **Sundaramurthy Pandurangan**, Neural network based prediction of less side effect causing cancer drug targets in the network of MAPK pathways**. International Journal of Bioinformatics Research and Applications.** 17(1): 69 – 79, 2021

2. MD AksamV.K, V. M. Chandrasekaran, and **Sundaramurthy Pandurangan ,**Protein Domain Level Cancer Drug Targets in the Network of MAPK pathways, **IEEE / ACM Transactions on Computational Biology and Bioinformatics,** 16(6):2057 - 2065, 2019.

3. MD Aksam V.K, Chandrasekaran V. M., and **Sundaramurthy Pandurangan,** Tological alternate centrality measure capturing drug targets in the network of MAPK pathways.**IET Systems Biology**12(5):226-232, 2018.

4. V.K. MD Aksam, V. M. Chandrasekaran, and **Sundaramurthy Pandurangan ,**Cancer drug target identification and node level analysis of the network of MAPK pathways.**Network Modeling Analysis in Health Informatics and Bioinformatics**, 7:4, 2018.

5. V. K. MD Aksam, V. M. Chandrasekaran, **Sundaramurthy Pandurangan**, Hub nodes in the network of human Mitogen-Activated Protein Kinase (MAPK) pathways: Characteristics and potential as drug targets.**Informatics in Medicine Unlocked**9, 173 – 180, 2017.

6.V.K. MD Aksam, V.M. Chandrasekaran, **Sundaramurthy Pandurangan**, Identification of cluster of proteins in the network of MAPK pathways as cancer drug targets. **Informatics in Medicine Unlocked**9, 86 – 92, 2017.

7.Appala Raju Kotaru, Khader Shameer, **Pandurangan Sundaramurthy,** Ramesh Chandra Joshi, An Improved Hypergeometric Probability Method for Identification of Functionally Linked Proteins using Phylogenetic Profiles. **Bioinformation,**9(7): 368-374, 2013.

8. Pandurangan Sundaramurthy, Raashi Srinivasan, Khader Shameer, Sunita Gakkhar, Ramanathan Sowdhamini. HORIBALFRE program: Higher Order Residue Interactions Based ALgorithm for Fold REcognition. **Bioinformation**7(7): 352-359, 2011.

9. Sundaramurthy Pandurangan, Shameer Khader, Raashi Srinivasan, Sunita Gakkhar, Sowdhamini Ramanathan. A bioinformatics protocol for the identification of spatial clusters and the calculation of higher order residue interactions in protein structures.**Protocol Exchange (2010)** http://www.nature.com/protocolexchange/protocols/655

10. Pandurangan Sundaramurthy, Khader Shameer, Raashi Srinivasan, Sunita Gakkhar, Ramanathan Sowdhamini. HORI: Higher Order Residue Interactions, a web server to compute possible interactions in a protein structure. **BMC Bioinformatics,** 11(Suppl 1):S24, 2010.

11. **Sundaramurthy P,** Gakkhar S. Dynamic modeling and simulation of JNK, and P38 kinase cascades with feedbacks and cross – talks. **IEEE Trans.Nanobioscience,** 9(4):225-31, 2010.

12. Sundaramurthy Pandurangan, Sunita Gakkhar. Systems Biology: To Meet Challenges for Biology in the Twenty-first Century, **Research in Biotechnology,** 1: 01 – 05, 2010 (Review)

13. Sundaramurthy Pandurangan, Sunita Gakkhar. Lose and gain: Impacts of ERK5 and JNK cascades on each other. **Systems and Synthetic Biology,** 4(2), 125-132, 2010

14. Pandurangan Sundaramurthy, Sunita Gakkhar, Ramanathan Sowdhamini. Analysis of ERK5, JNK, and P38 kinase cascades:A systems approach,**Bioinformation,**3(6),250–254, 2009.

15. Pandurangan Sundaramurthy, Sunita Gakkhar, Ramanathan Sowdhamini. Computational prediction and analysis of impact of the cross talks between JNK and P38 kinase signaling pathways.**Bioinformation,** 3(6), 244 – 249, 2009.

ORAL / POSTER PRESENTATIONS

1. Neural network based prediction of less side effects causing cancer drug targets in the network of MAPK pathways, 7–9 Nov. 2017, Inbix'17: Indian Conference on Bioinformatics 2017, Birla Institute of Scientific Research, Jaipur.

2. Identification of cluster of proteins in the network of MAPK pathways as cancer drug targets, 28- 30, Nov. 2016, International Conference on Advances in Scientific Computing (ICASC 2016), Indian Institute of Technology Madras.

3. Characterization and implications of targeting hub nodes in the network of MAPK pathways,23 - 26, Nov. 2015, International Conference on Systems Biology 2015 (ICSB2015), Singapore.

4. HORIBALFRE: Higher Order Residue Interactions Based Algorithm for Fold Recognition. Feb. 3 - 5, 2011, 1st IEEE International Conference on Computational Advances in Bio and medical Sciences (ICCABS), Florida.

5. HORI: a web server to compute Higher Order Residue Interactions in protein structures, 18 – 21 Jan. 2010, The Eighth Asia Pacific Bioinformatics Conference (APBC 2010), Bangalore.

6. Prediction and Analysis of the Dynamical Behaviors of ERK5, JNK and P38 Kinase Cascades, 07 Nov. 2009, Ph.D. thesis viva presentation, Department of Mathematics, Indian Institute of Technology Roorkee.

7. Bioinformatics – A Multidisciplinary Applied Science, 21-27 Nov. 2005, Research Level Group Discussion on Natural Computing, Indian Institute of Technology Madras.

8. Structure Based Drug Design – Highly Challenging and Rewarding, 21-27 Nov. 2005, Research Level Group Discussion on Natural Computing, Indian Institute of Technology Madras.

9. Structure Based Strategies for Drug Design, 08 June 2005, Comprehensive Oral Presentation, Department of Mathematics, Indian Institute of Technology Roorkee.

INVITED TALKS

1. Quantitative Analysis of Interactions among the Kinase Cascades in MAPK Pathways. 04th November 2016, Department of Life Sciences, Central University of Tamil Nadu, Thiruvarur.

2. 21st Century Biology: Omics, Systems and Networks. 16th March 2016, National Conference on*"Current Scenario in Biotechnology*", Department of Biotechnology, Kumaraguru College of Technology, Coimbatore.

3. Dynamical behaviors of cancer signaling networks. 14 September 2013, Department of Mathematics, Sacred Heart College, Tirupattur.

4. Linear Program for a grand challenging problem in bioinformatics. 2-5 July 2012, FDP on Probability and Statistics for Engineers,School of Advanced Sciences, Vellore Institute of Technology, Vellore Campus.

5. Systems Biology of Signaling Networks. 18 October 2011, School of Biotechnology, Madurai Kamaraj University.

6. Mathematics for grand challenging problems in biology. 5-7 July 2011, FDP on Mathematics and Statistics for Biologists,School of Bio Sciences and Technology, Vellore Institute of Technology, Vellore Campus.

7. Is it right to correlate a signaling pathway to cancer?,1 July 2011, School of Bio Sciences and Technology, Vellore Institute of Technology, Vellore Campus.

8. Network Based Drug Discovery 14 – 15 May 2011, National Level Seminor on Current Trends and Advanced Analytical Procedures in Pharma Industry. The Oxford College of Pharmacy, Bangalore.

9. Mathematical Flavor on Bioinformatics – I & II, 07 – 14 July 2010, Instructional Workshop on Mathematical Modeling and Simulation, National Center for Advanced Research in Discrete Mathematics (n-CARMATH), Kalasalingam University, Krishnankoil.

10. On the Dynamic Pathway Modeling and its Applications, 12 -13 March 2009, National Conference on Applications of Mathematics in Science and Technology, Sacred Heart College, Tirupattur.

11. Bioinformatics: Who and What Can Do?, 17 Sep. 2007, KCG College of Technology, Chennai.

12. Bioinformatics: Opportunities and Challenges, 06 April 2006, Rajalakshmi Engineering College, Chennai.

CONFERENCES / WORKSHOPS / SYMPOSIUMS/ SEMINORS / MEETUPSATTENDED

1. Amazon AI Conclave (Alexa Skills Workshop),19th December 2019, Amazon Internet Services Pvt. Ltd,Bangalore.

2. Business Comes Alive (Conference on IoT, AI and 5G),22nd October 2019, Intel Corporation, Bangalore.

3. New Frontiers in Deep Learning (Meetup),26th May 2018, Sapient India, Bangalore.

4. Artificial Intelligence (Bootcamp), 7 – 8th April 2018, Great Learning, Bangalore.

5. Powering the Future with Analytics, 22nd July 2017, Accenture's Analytics Forum,Bangalore.

6. Data Platform Day, 18th March 2017, Microsoft, Bangalore.

7. Artificial Intelligence Seminar,9th Dec. 2016, Intel® Software, Bangalore.

8. Computational Cell Biology, March 28 – April 1, 2011. Cold Spring Harbor Laboratory, New York.

9. Systems Biology: Networks, March 22-26, 2011.Cold Spring Harbor Laboratory, New York.

10. **Advances in Genome Biology and Technology (AGBT),** February 2-5, 2011. Marco Island, Florida.

11. Grant Writing Workshop, 09 – 10 Dec. 2010, Cold Spring Harbor Laboratory, New York.

12. Personal Genomes, 10 – 12 Sep. 2010, Cold Spring Harbor Laboratory, New York.

13. International Symposium on Interface between Chemistry and Biology (CHEMBIO)**,** 17-18 Feb. 2009, Indian Institute of Science, Bangalore and 19 Feb. 2009, National Center for Biological Sciences, Bangalore.

14. Third Indo – French Bioinformatics Meeting (IFBM), 12-14 June 2006. National Center for Biological Sciences, Bangalore.

15. 71ST Annual Conference of Indian Mathematical Society (IMS), 26-29 Dec. 2005, Department of Mathematics, Indian Institute of Technology Roorkee.

16. Machine Learning Techniques in Functional Proteomics, 18-21 Oct. 2005, Institute of Microbial Technology, Chandigarh.

17. Symposium on Current Trends in Biomathematics, 11-13 March 2005. Department of Mathematics, Indian Institute of Technology Roorkee.

18. Drug Discovery and Development in New Millennium -2 (D3NM – 2), Feb 28 – March 03 2005, National Institute of Pharmaceutical Education and Research, Chandigarh.